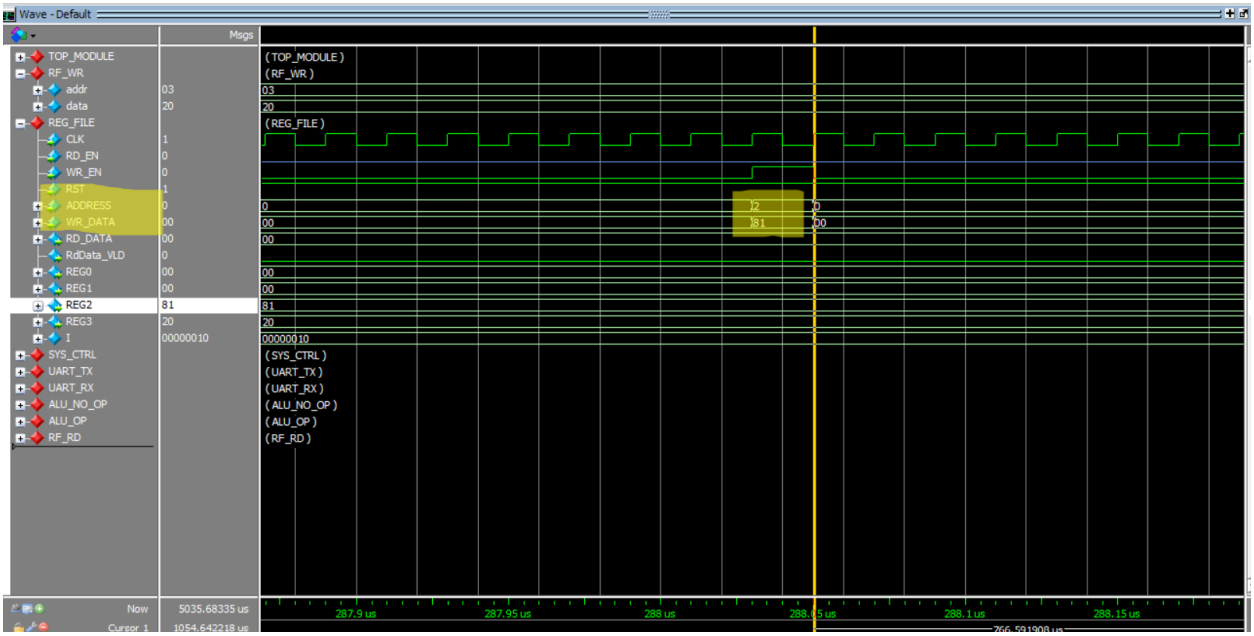
FINAL_SYSYTEM
PHASE 1

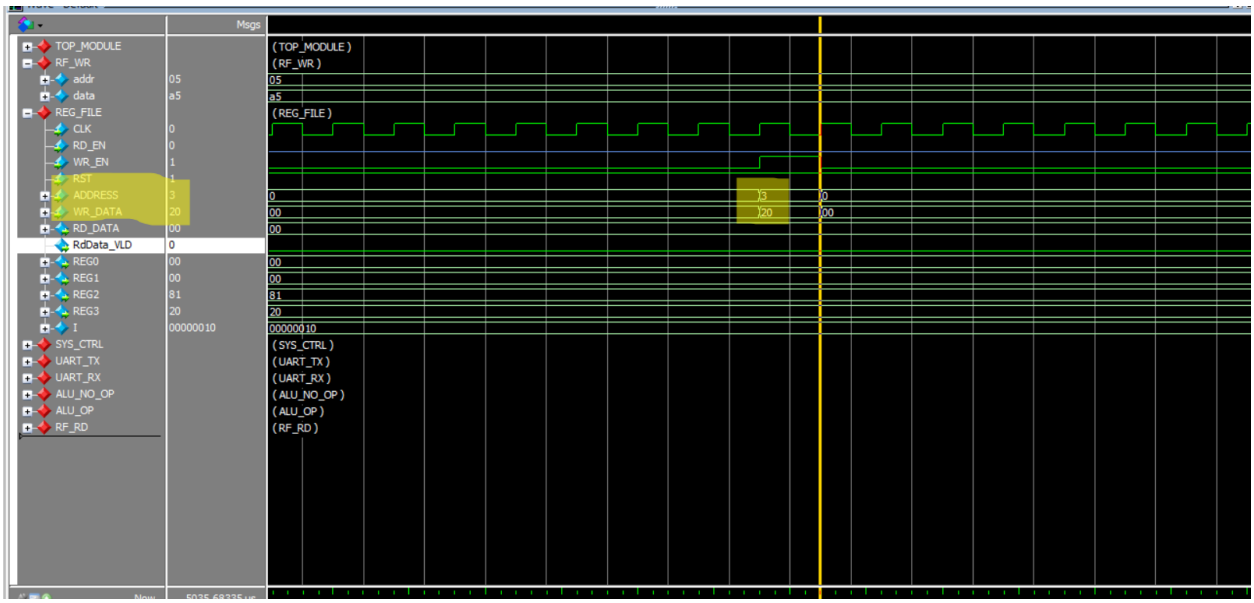
First I show here the transcripts showing all tests pass and then snippests of the code showing 4 test cases.

```
# Executing ONERROR command at macro ./wave.do line 14
# TEST 1: configuration (REG2, REG3)
# TEST 2: RF write then read back
#   PASS: got 0x00a5
#   PASS: got 0x003c
# TEST 3: ALU with operands (A = 5, B = 3)  ADD-SUB-MULT-DIV
#   PASS: got 0x0008
#   PASS: got 0x0002
#   PASS: got 0x000f
#   PASS: got 0x0002
# TEST 4: ALU without operands (A = 6, B = 3 still in REG0/REG1)  ADD-SUB
#   PASS: got 0x0009
#   PASS: got 0x0003
# TEST 5: UART error flags
#   PASS: no parity/framing errors
# ALL TESTS PASSED
# ** Note: $finish    : ../Test_Bench/SYSTEM_tb.v(211)
#    Time: 5035683350 ps  Iteration: 0  Instance: /SYS_TOP_tb
# 1
```

Breakdown to configure reg 2 and reg 3:



as we se here reg at address 2 is configered with 0xh 81 and reg 3 with 0xh 20

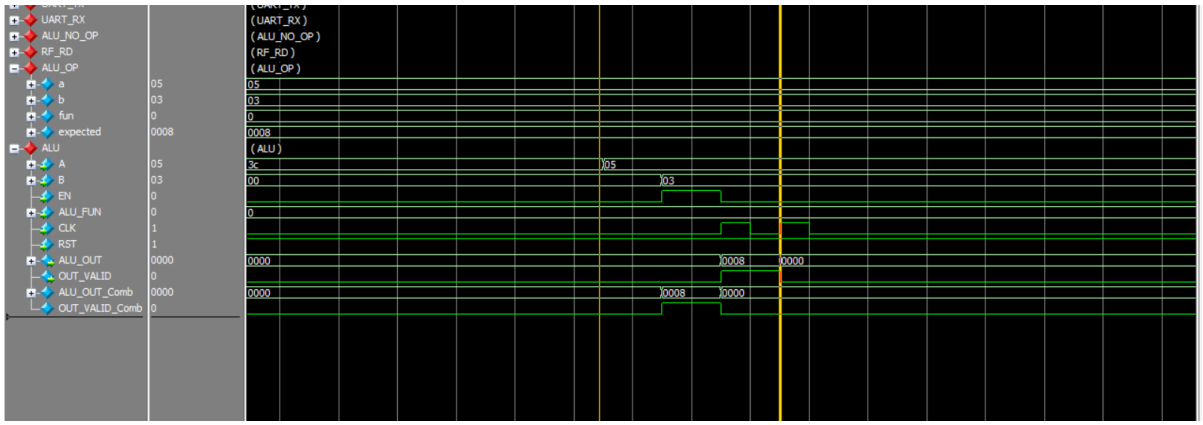


NEXT, we move to ALU operations:

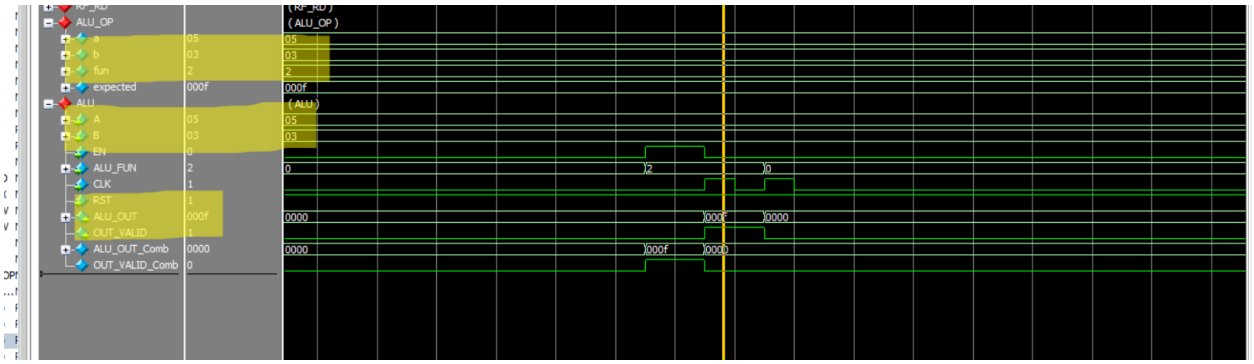
ALU ADDITION with operands

Code snippest showing tasks performed to test ALU

```
// ------------------------------------------------------------------
$display("TEST 3: ALU with operands (A = 5, B = 3)  ADD-SUB-MULT-DIV");
alu_with_op(8'd5, 8'd3, ADD, 16'd8);
alu_with_op(8'd5, 8'd3, SUB, 16'd2);
alu_with_op(8'd5, 8'd3, MUL, 16'd15);
alu_with_op(8'd6, 8'd3, DIV, 16'd2);
```



Here I have ALU_OP task and the unit ALU and the matches the same result of adding a,b.



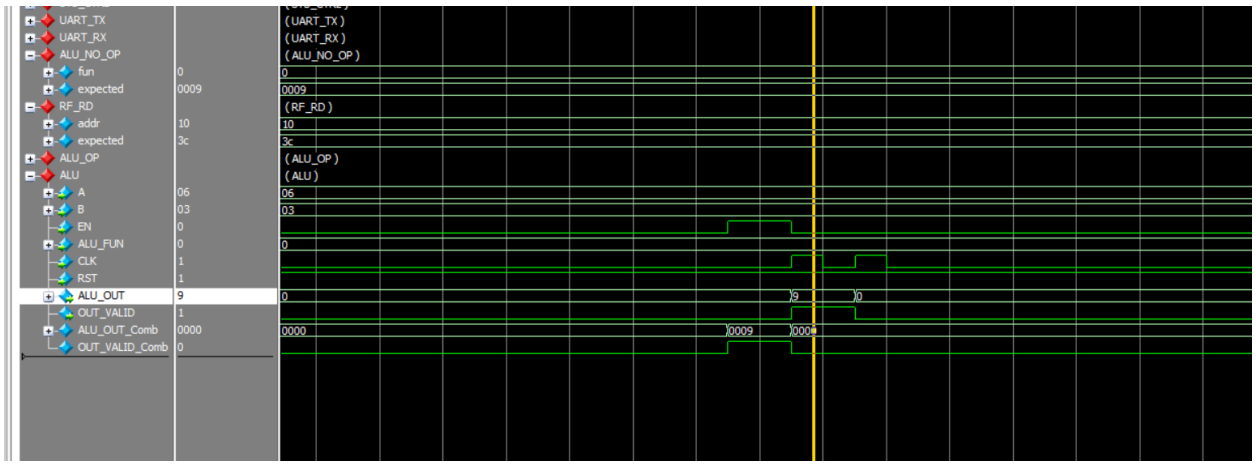
And here multiplication which lead to result of 15 => pased;

ALU ADDITION with no operands

Code snippest showing tasks performed to test ALU

```
// ------------------------------------------------------------------
$display("TEST 4: ALU without operands (A = 6, B = 3 still in REG0/REG1)  ADD-SUB");
alu_no_op(ADD, 16'd9);
alu_no_op(SUB, 16'd3);

//
```



here is addition with no operands and we got the result of 9 => pass;